

Microbial Enzymes

Enzymes in industry

Enzymes have several advantages over chemical catalysts:

1. The ability to function under relatively mild conditions of temperature, pH and pressure.
2. Enzymes are specific, often stereoselective, catalysts, which do not produce unwanted byproducts.
3. Enzyme-based processes are 'environmentally friendly'
4. Certain enzymes are not restricted to aqueous environments and can operate in two-phase water-organic solvent systems and in non-aqueous organic media, particularly hydrophobic solvents.

* enzymes are recyclable (can be reused)

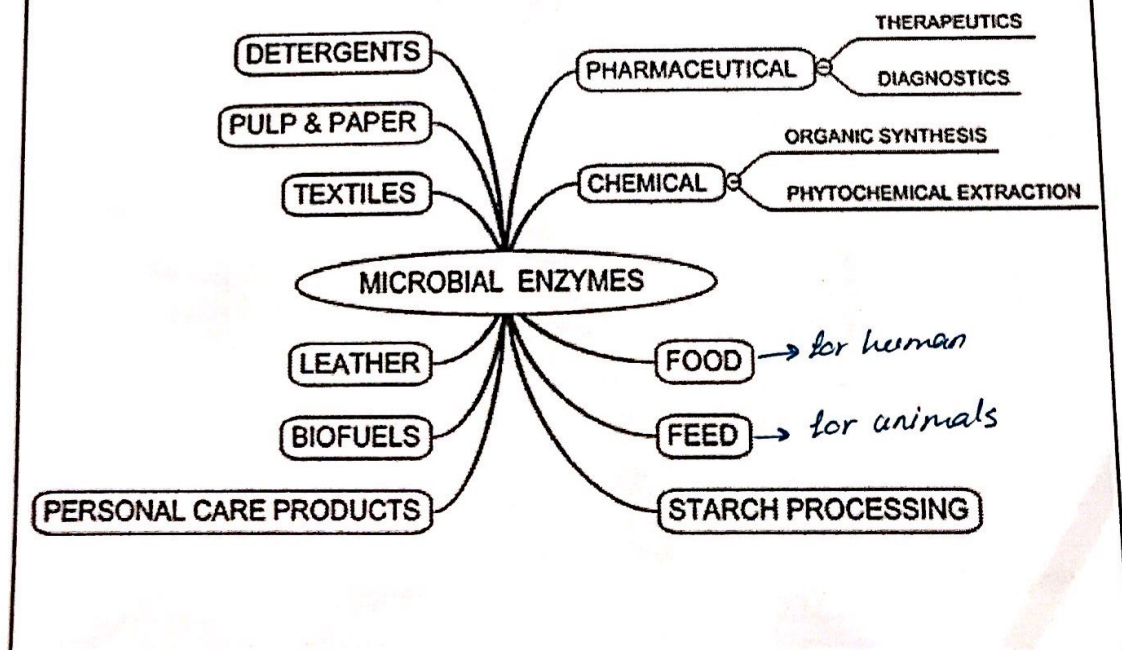
Enzymes classification

Table 9.1 Classification of enzymes based on the types of reaction catalyzed

Class	Name	Type of reaction	Examples
1	<u>Oxido-reductases</u>	<u>Oxidation- reduction reactions</u> , generally require NADP/NAD as cofactors	Lactate dehydrogenase
2	<u>Transferases</u>	<u>Transfer of functional groups</u> like phosphate, acetyl or amino groups	Hexokinase
3	<u>Hydrolases</u>	<u>Hydrolysis reactions</u> , i.e. addition of water	α -amylase
4	<u>Lyases</u>	<u>Addition or removal of groups to form double bonds</u>	Pyruvate decarboxylase
5	<u>Isomerases</u>	<u>Rearrangement of atoms within a molecule</u>	Methionine racemase
6	<u>Ligases</u>	<u>Joining of two substrates by expense of ATP hydrolysis</u>	DNA synthetase

bio

Enzymes Applications



Bulk vs Fine microbial enzymes

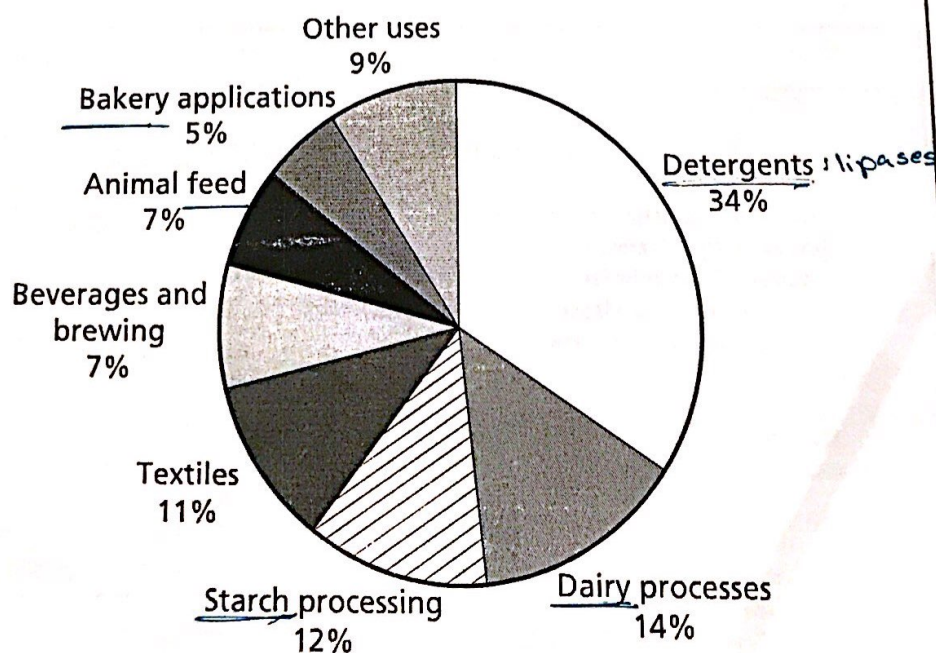
Bulk Enzymes preparations

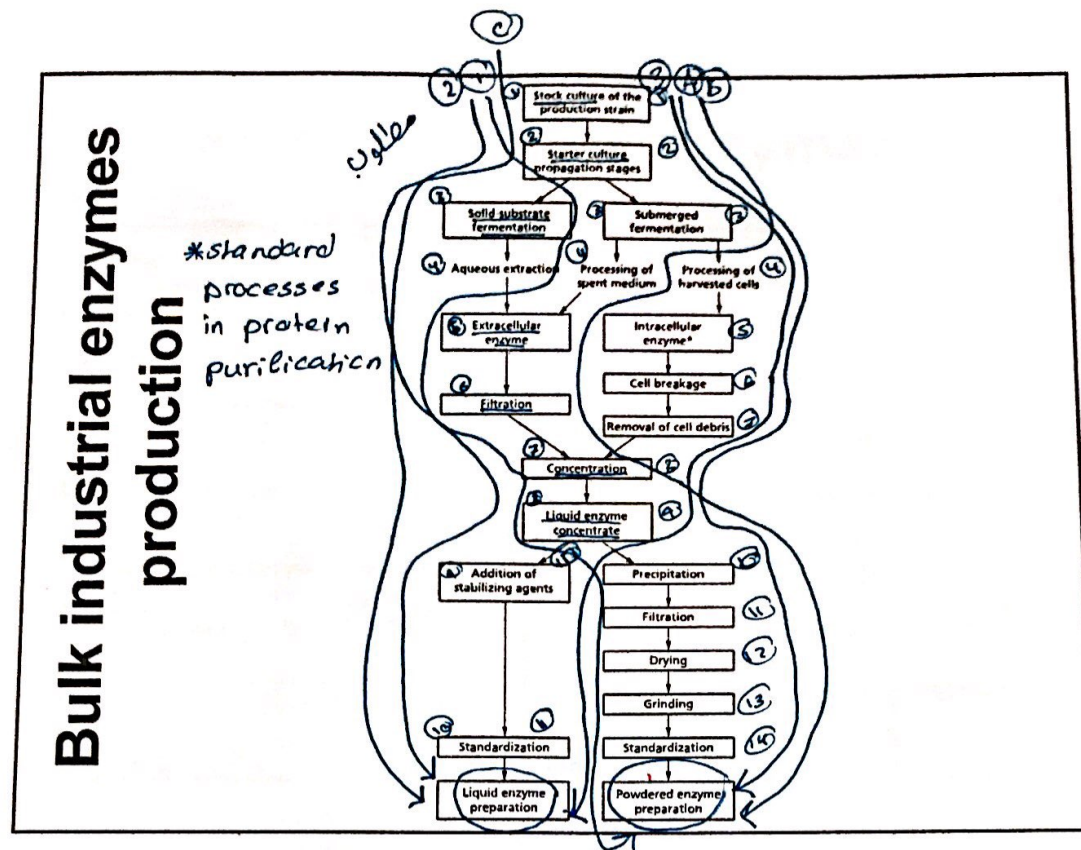
- Bulk enzyme are ^① partially purified and ^② produced in large quantities
- Bulk enzyme preparations are preferred for numerous and varied industrial processes.

Fine Enzyme preparations

- Produced in ^① smaller quantities, we need them for certain fine chemicals
- ^② High-purity enzyme preparations : for use as a drug
- Required for numerous applications, such as ^③ therapeutic agents, many of which are now recombinant products

Bulk enzymes





Fine microbial enzymes

Enzyme	Use
① <u>α-Amylase</u> EC 3.2.1.1	① Digestive aids
② <u>Alcohol dehydrogenases</u> EC 1.1.1.1 or <u>alcohol oxidases</u> EC 1.1.3.13	② Ethanol testing
③ <u>Asparaginase</u> EC 3.5.1.1	③ Maintains low asparagine levels in treating cancers (lymphomas and leukaemias) whose cells cannot synthesize the amino acid
④ <u>Catalase</u> EC 1.11.1.6	④ Contact lens cleaning systems
⑤ <u>Cholesterol esterase</u> EC 3.1.1.13	⑤ Monitoring serum cholesterol levels
⑥ <u>Creatininase</u> EC 3.5.2.10 (creatinine amidohydrolase)	⑥ Determination of serum creatinine levels
⑦ <u>Glucose oxidase</u> EC 1.1.3.4	⑦ Analysis of blood glucose levels
⑧ <u>β-Lactamase</u> EC 3.5.2.6	⑧ Treatment of penicillin allergy
⑨ <u>Proteases</u> (various)	⑨ Used as digestive aids, for wound debridement and contact lens cleaning

removing br: dements

Fine microbial enzymes

Enzyme	Use
⑩ Rhodanase EC 2.8.1.1 (thiosulphate sulphur transferase)	⑩ Treatment for cyanide poisoning
⑪ Streptokinase EC 3.4.22.10 (a cysteine protease)	⑪ To break down blood clots
⑫ Urease EC 3.5.1.5	⑫ Removal of urea from blood in renal failure
⑬ Uricase EC 1.7.3.3 (urate oxidase)	⑬ Used in the treatment and diagnosis of gout → النقرص
Some microbial enzymes used in molecular biology (*DNA directed enzymes)	
① DNA ligases (e.g. ATP, EC 6.5.1.1)	① Joining pieces of DNA
② DNA polymerase* EC 2.7.7.7	② DNA synthesis, e.g. in PCR for forensic analysis
③ Restriction endonucleases e.g. <i>Bam</i> HI and <i>Eco</i> RI	③ Cutting DNA at specific base sequences
④ RNA polymerase* EC 2.7.7.6	④ RNA synthesis

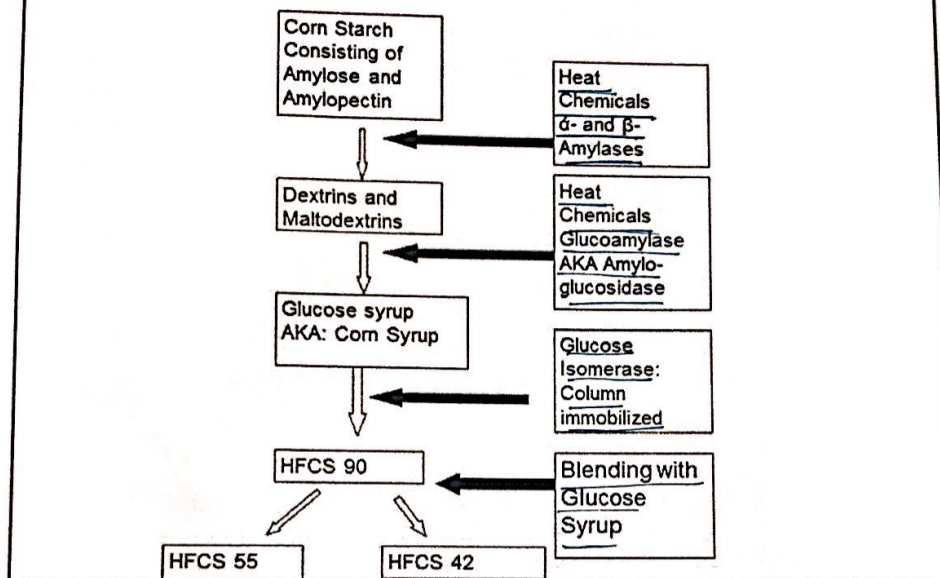
enzymes that are used to break blood clots within 2-3 hours, from streptococcus

Food & feed industry

<u>Amylases</u>		<u>Lipases</u>	Oils and fats, baking, and dairy industry
<u>α-amylase</u>	Starch syrups, fermentation and animal feed	<u>Naringinases</u>	Debittering of citrus juice and peel → removing the bitter taste from food
<u>β-amylase</u>	Brewing, maltose syrups	<u>Pectinase</u>	Fruits and fruit juice processing
<u>Glucoamylase</u>	Manufacture of dextrose syrups and HFCS	<u>Phytases</u>	Fish feeds
<u>α-galactosidase</u>	Enhances sucrose yield	<u>Tannases</u>	Reducing undesirable effects of tannins → in tea, phenolic compounds, bind to iron and make it unavailable
<u>Cellulases</u>		<u>Transglutaminases</u>	Texture improvement of various kinds of noodles, pasta, hams and sausages, crumb strength
<u>β-glucanase</u>	Animal feed, extraction and clarification of fruit and vegetable juices brewing industry		
<u>β-glucosidase</u>	Transforms isoflavone phytoestrogens in soymilk		
<u>Dextranase</u>	Hydrolysis of the polysaccharide dextran		
<u>Invertase</u>	Invert syrup of cane or beet sugar		
<u>Keratinases</u>	Commercial feather meal for poultry and fish feed		
<u>Lactase</u>	Metabolizes/reduces lactase from dairy foods		

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High Fructose Corn Syrup (HFCS)



Detergent industry

1. Proteases ✓
2. Lipases ✓ *lipids are the hardest to be removed
(removed by lipases)*
3. Cellulases ✓
4. Amylases ✓
5. Mannanases ✓

Textile Industry

Desizing

- Prior to weaving fabrics, the threads are reinforced with an adhesive (termed 'size') to prevent breakage of warp threads.
- The size may be composed of starch or cellulose derivatives
- Before dyeing, bleaching and printing, the size is removed from the cloth with amylases

Bioscouring

- Untreated cotton contains varieties of cellulosic impurities like waxes, pectins, and hemicelluloses, which give hydrophobic properties to the fiber and thereby interfere in the process of dyeing and finishing.
- Microbial enzymes like pectinases, proteases, lipases, and cutinases remove these impurities and enhance wettability of the fiber.

→ cotton has certain impurities
→ decreasing hydrophobicity (making it hydrophilic)

Textile Industry

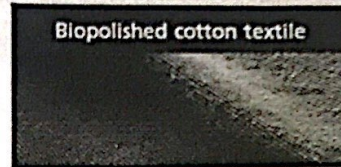
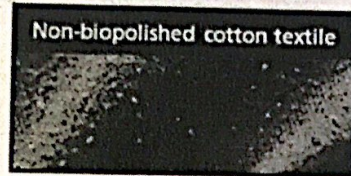
Biopolishing

- Cellulases are also employed in bio-polishing cotton and other cellulose fibres to produce fabrics with a smoother and glossier appearance

Denim washing

- Treatment of denim garments to give a worn look before sale is called 'stonewashing'.
- Cellulases are now used to accelerate abrasion and aid the loosening of the indigo dye.

Textile Industry



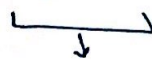
Leather industry

The most important applications of microbial proteases and lipases during:

1. Soaking
2. De-hairing
3. De-greasing
4. Bating

the most expensive leathers:

lamb - cows - pigs



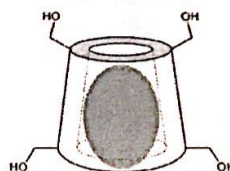
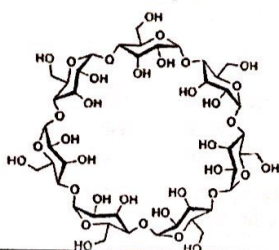
smooth



contain pores

Enzymes as catalysts in organic synthesis

- Microbial enzymes are used in the synthesis of high-value organic compounds for the chemical, food and pharmaceutical industries.
- For example, cyclodextrin glycosyltransferases, create cyclodextrins from simple starch molecules.
- Cyclodextrins are useful carriers of fragile compounds such as drugs, vitamins and flavours.
- No chemical means exists for creating this class of substances, so the enzymatic route is unique.



a cyclic molecules
used to put a certain
material inside to
protect it from
stomach acids

cyclodextrins
→ drug

Enzymes as catalysts in organic synthesis

- An increasingly important field is the synthesis of chiral compounds. *same structure but different arrangements*
- Compounds are chemically synthesized an almost 50 : 50 mixture of enantiomers is made, which is called a racemic mixture.
- In the industrial synthesis of chiral compounds, there have to be controls to guarantee the production of only the desired enantiomer, often at a purity higher than 99%.
- Enzyme-based processes can be used to prepare specific enantiomers and resolve enantiomers from racemic mixtures.
- These processes utilize racemases, which are a family of enzymes that accept either enantiomer, and function to turn a substrate of one chirality into its opposite form.
(like transferring glucose to galactose)

Pulp and Paper Processing

- Paper is composed of natural biopolymers: cellulose, hemicellulose, and lignin.
- Papermaking involves the use of chemicals to recover the cellulosic fibers.

Table 9.18 Milestones in the application of enzymes in the pulp and paper industry

Year	Enzyme used	Process
1959	Cellulase	Pulp fibrillation
1984	Xylanases	Enzymatic beating and hemicelluloses removal
1986	Xylanases	Pre-bleaching
1989	Lipase	Pulp depitching
1993	Laccases	Pulp de-lignification
1996	Manganese peroxidase	Bleaching

Biofuel

- Biofuels are defined as fuels derived from biomass conversion such as biodiesel, bioethanol, biohydrogen, and biogas.
↳ can be extracted from organic waste
- Biodiesel (mono alkyl ester) of long-chain of fatty acid has immense potential as an alternative fuel
- Lipase (Biodiesel) & Cellulase (Biobutanol, Bioethanol)

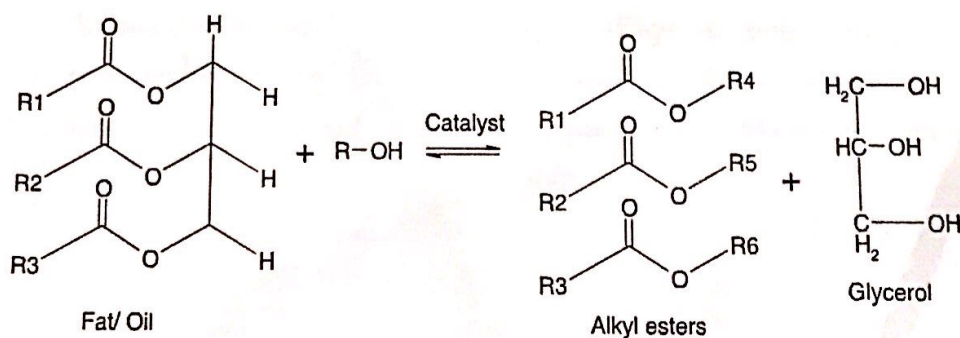


Fig. 9.4 Transesterification process

Enzymes immobilization

Advantages of enzyme immobilization:

1. Reuse
2. Suitability for application within continuous operations
3. Their product is enzyme free, therefore further processing to remove or inactivate the enzyme is not required
4. Improved enzyme stability
5. Reduced effluent disposal problems.

Enzymes immobilization

